

มหาวิทยาลัยมหิดล
Mahidol University



ITCS215
Introduction to Interactive Multimedia Technology

Semester 1/2026

Introduction to Image Processing/Computer Vision

Assoc. Prof. Dr. Worapan Kusakunniran

3 Modules

Module 1 (5 weeks): Introduction to Image Analysis and Machine Learning.



Module 2 (5 weeks): Introduction to Computer Graphic, Generative AI, Sound-Visualization/Analytics.



Module 3 (5 weeks): Introduction to Animation, Interactive Application, Text and Font.



Module 3: Assessments

3 Parts

Quiz	30.00
Final Exam	40.00
Class Activities and Assignments	30.00
	100.00

- Class activities

  10 %
- Quiz

  10 %
- Final Exam

  40/3 ~ 13.33 %
- Bonus for module 1

  max = 2% to be topped up for the missing scores in class activities and quiz, but limited to 20% max (act+quiz+bonus)

Module 3: Topics

4 Parts

https://rb.gy/xu6bj6

1. Neighborhood operations and non-uniform image scaling (seam carving)
2. Color Theory and its applications
3. AI for image processing and related applications
4. Introduction to computer vision, its tool (OpenCV) and applications (i.e., motion analysis)

Prepare computer machine

- Group work (Max. 4 members)
  at least one laptop
- Deadline: Before next class (via myCourse)

Computer Vision

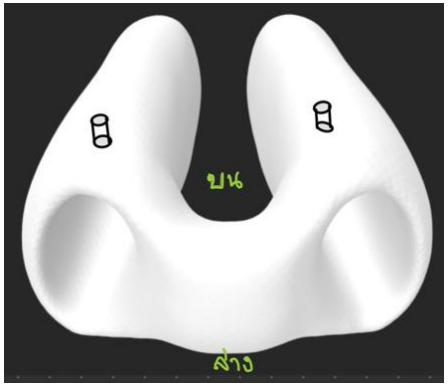
- Very broadly, computer vision is the science and technology of machines that see

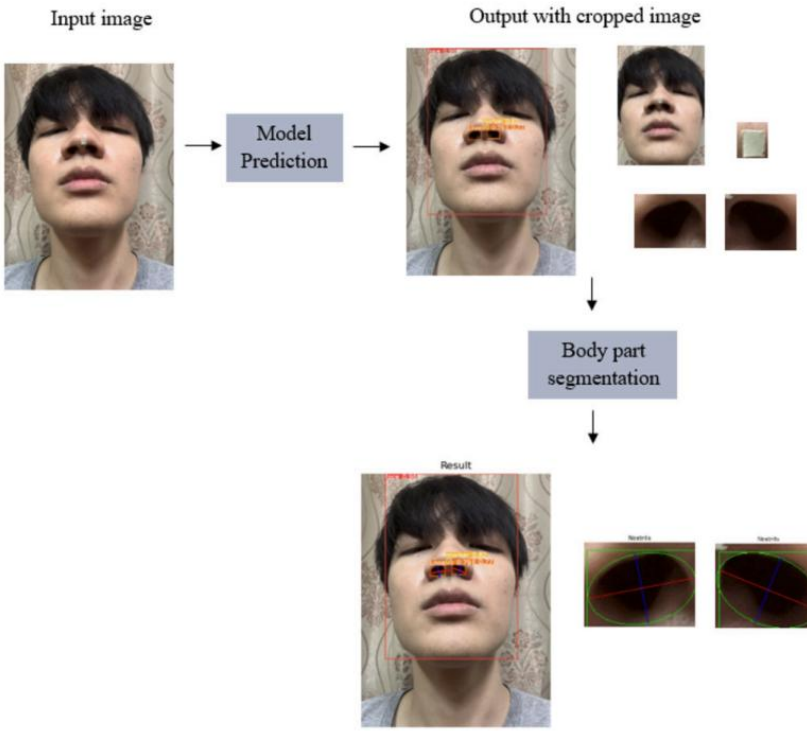
- This means taking in images or videos and extracting useful information from them

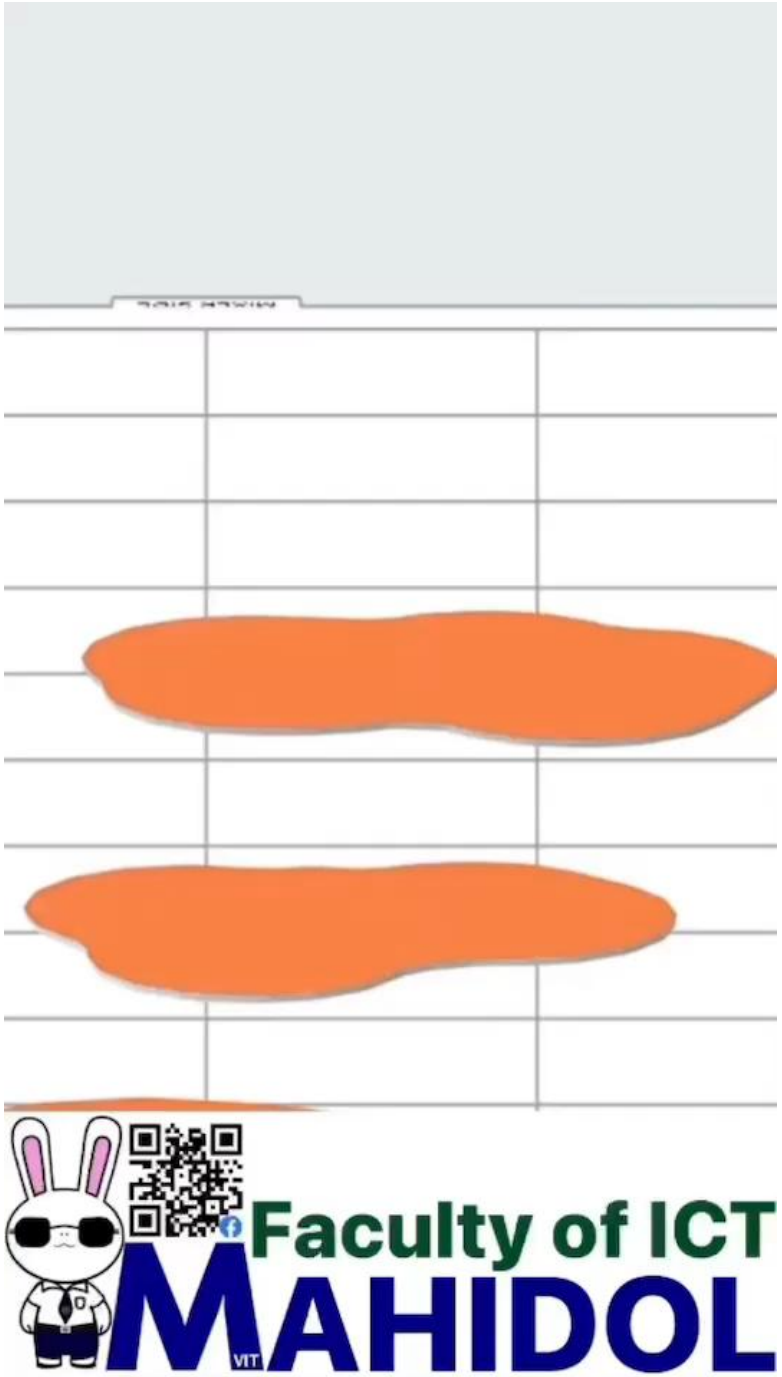
- Traditionally, computer vision is seen as the inverse of computer graphics

Nasal Splint



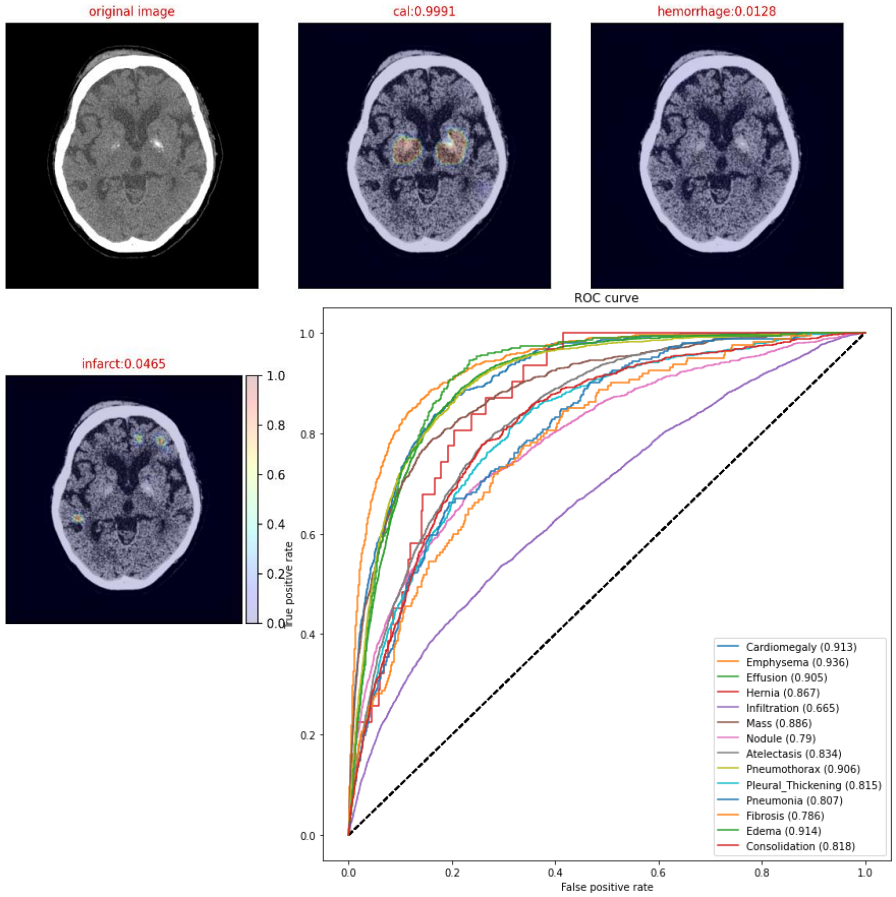


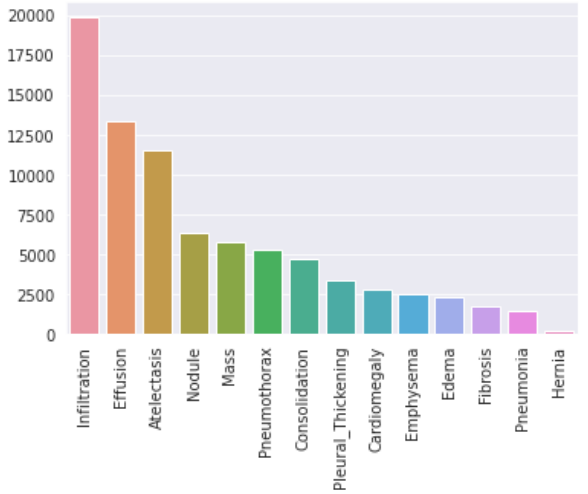


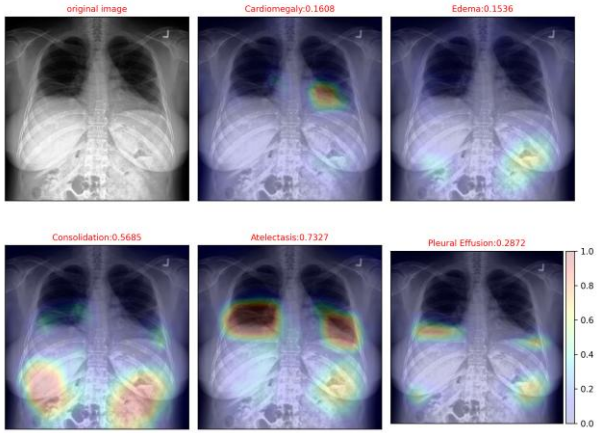


Faculty of ICT
MAHIDOL

Brain CT/ Chest PA





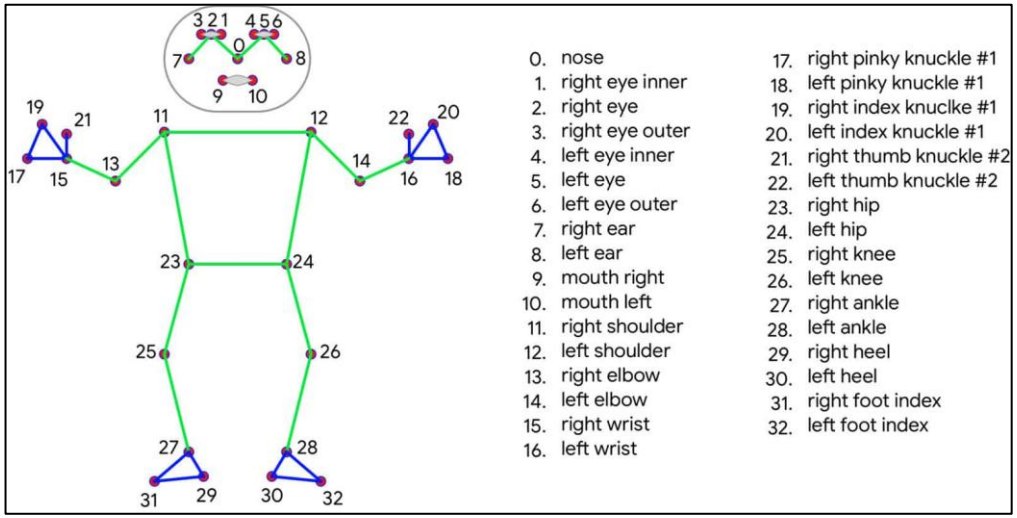


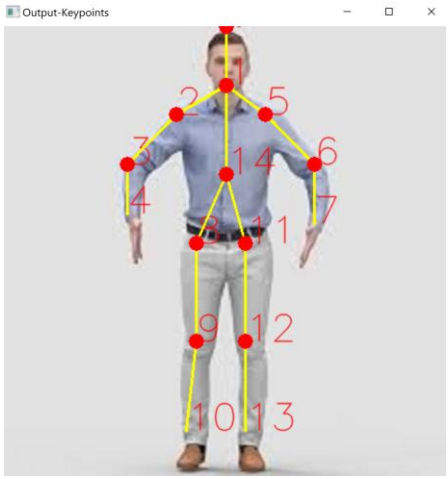
https://www.kaggle.com/nih-chest-xrays/data

DEMO

Posture/Key Joints

MediaPipe Pose (33 keypoints)





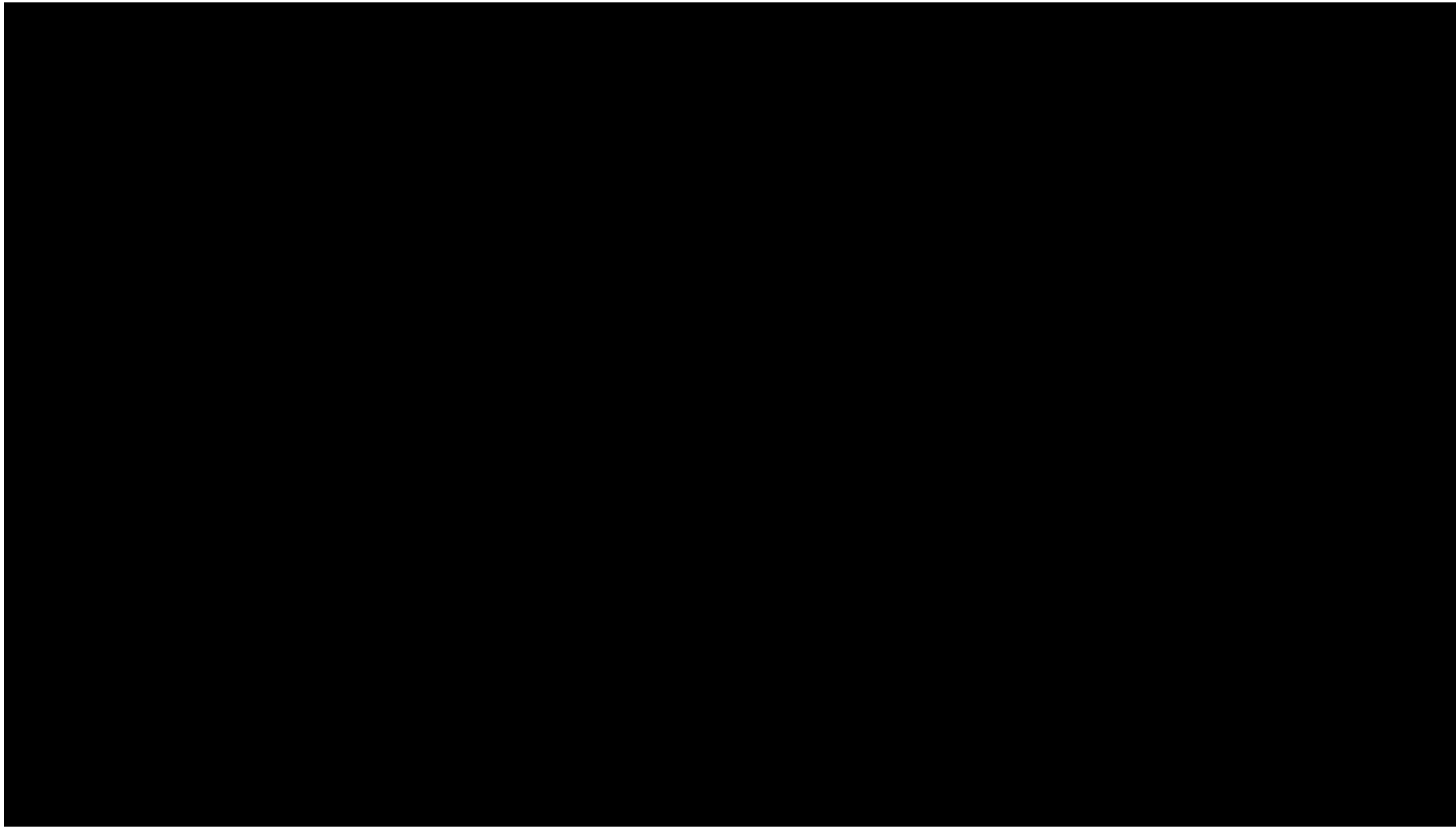
MPII Output Format
Head – 0
Neck – 1
Right Shoulder – 2
Right Elbow – 3
Right Wrist – 4
Left Shoulder – 5
Left Elbow – 6
Left Wrist – 7
Right Hip – 8
Right Knee – 9
Right Ankle – 10
Left Hip – 11
Left Knee – 12
Left Ankle – 13
Chest – 14
Background – 15

Coco/OpenPose
0 Nose
1 Left eye
2 Right eye
3 Left ear
4 Right ear
5 Left shoulder
6 Right shoulder
7 Left elbow
8 Right elbow
9 Left wrist
10 Right wrist
11 Left hip
12 Right hip
13 Left knee 1
4 Right knee
15 Left ankle
16 Right ankle

Motion Game/App

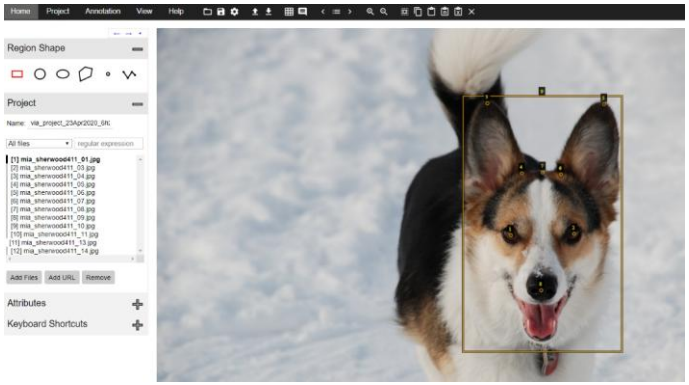


Motion Game/App

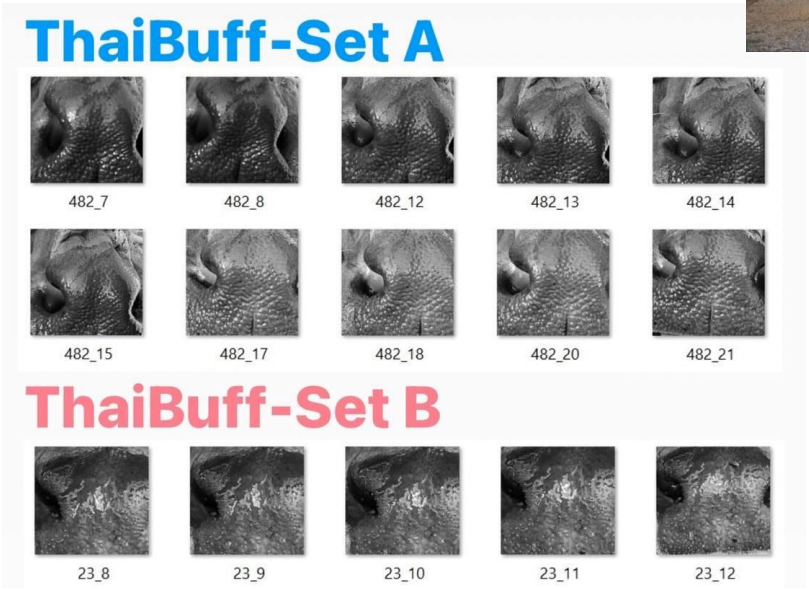


Animal Biometric

- Counting
- Re-id
- Breed classify









Cocoa Fruits



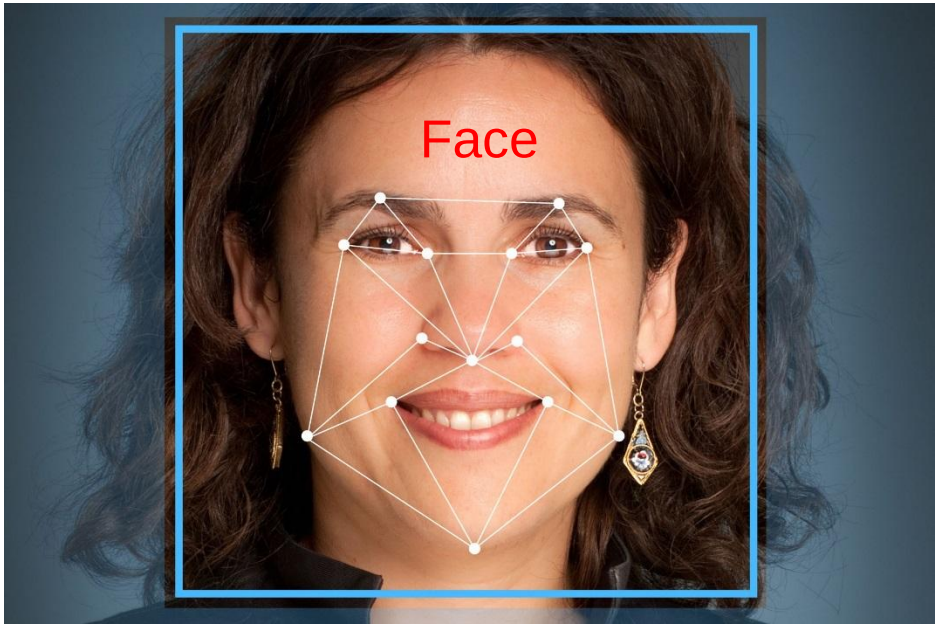
Mangosteen Grading







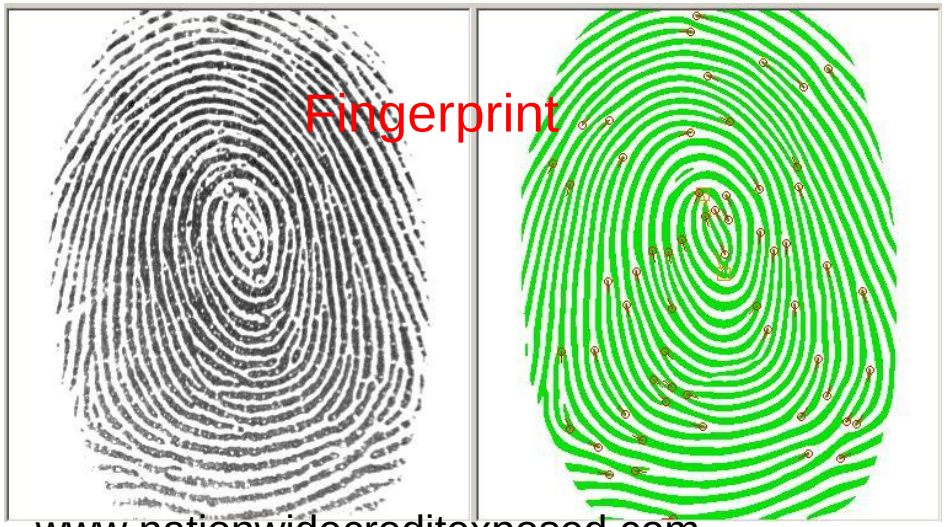
State Of The Art



www.wired.co.uk



www.brightsideofnews.info

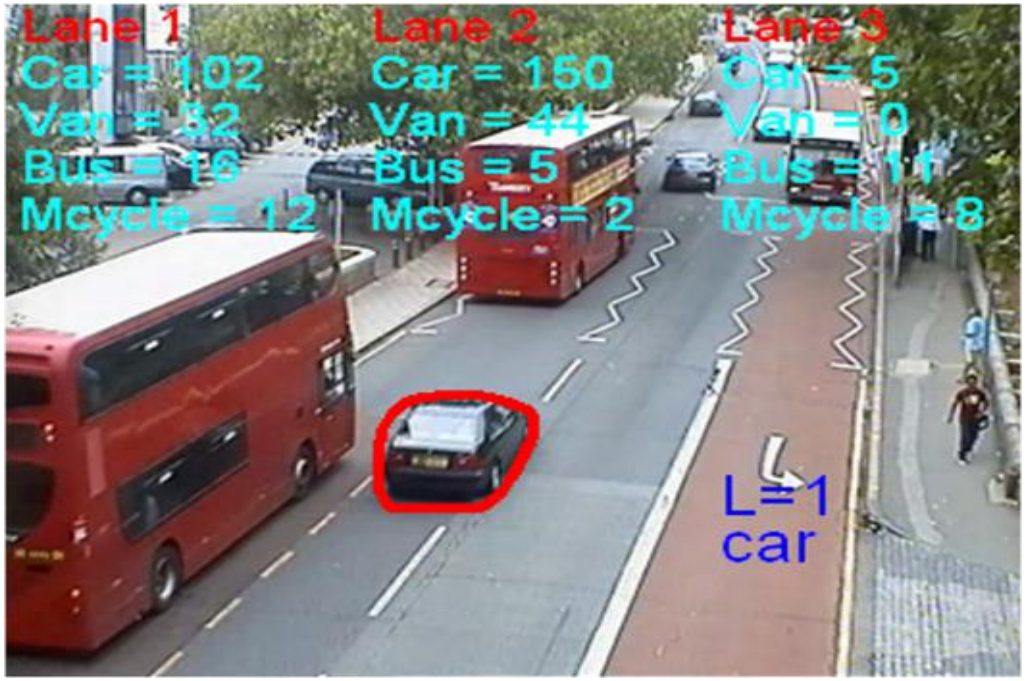


www.nationwidecreditexposed.com

Biometrics

Object Classification

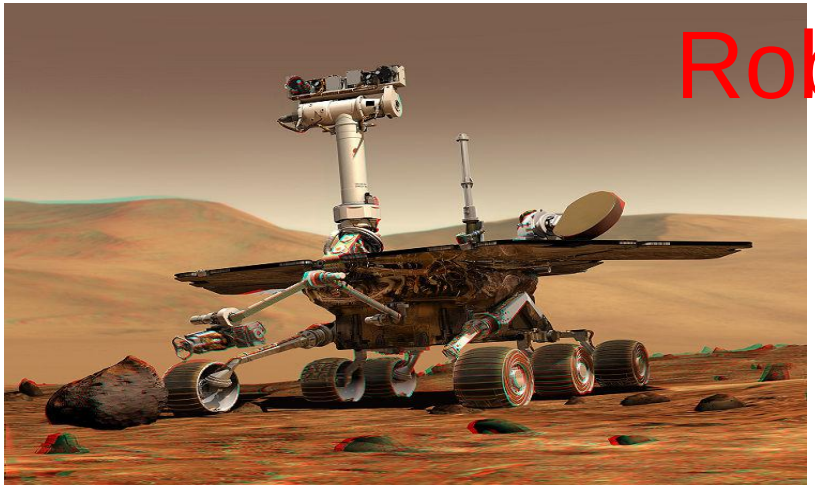
- Surveillance application
  - Pedestrian vs. Car vs. Van vs. bus vs. Motorcycle



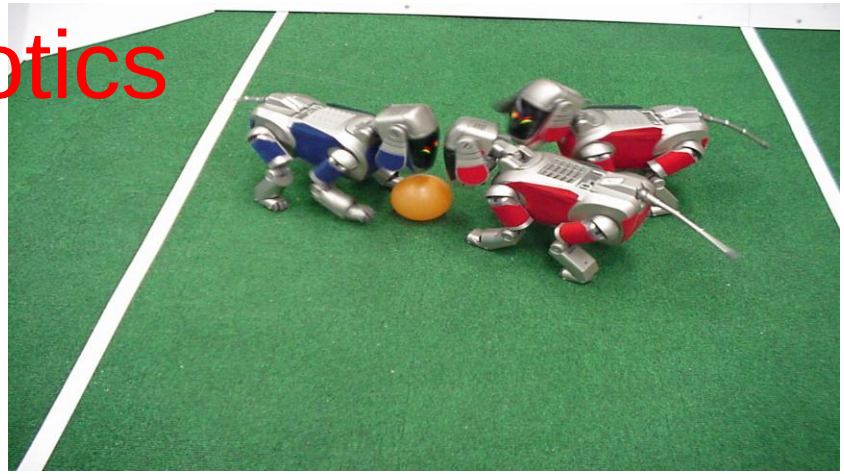
Chen Z, Ellis T, Velastin SA. Vision-based traffic surveys in urban environments. Journal of Electronic Imaging. 2016 Sep 1;25(5):051206-.

State Of The Art

Robotics



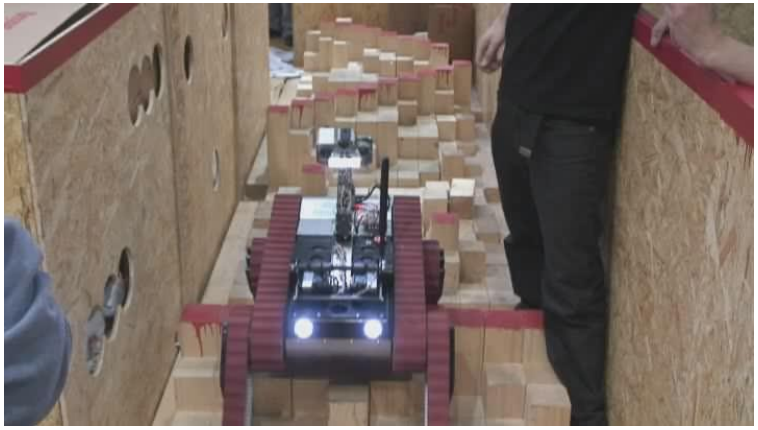
NASA’s Mars Spirit Rover
http://en.wikipedia.org/wiki/Spirit_rover



http://www.robocup.org/



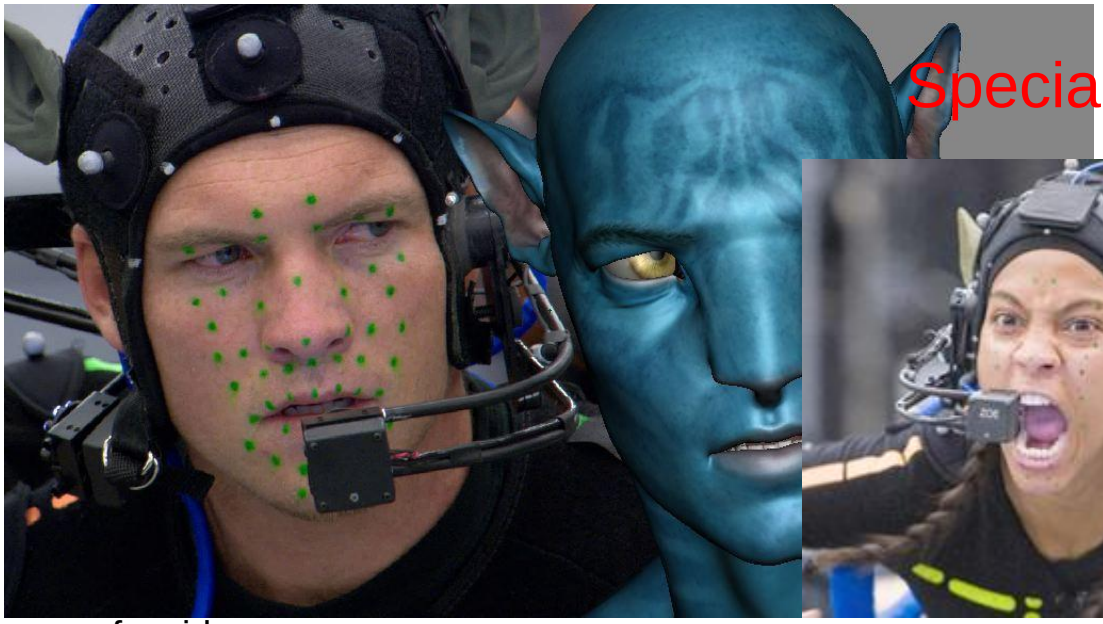
www.designboom.com



www.robaid.com

State Of The Art

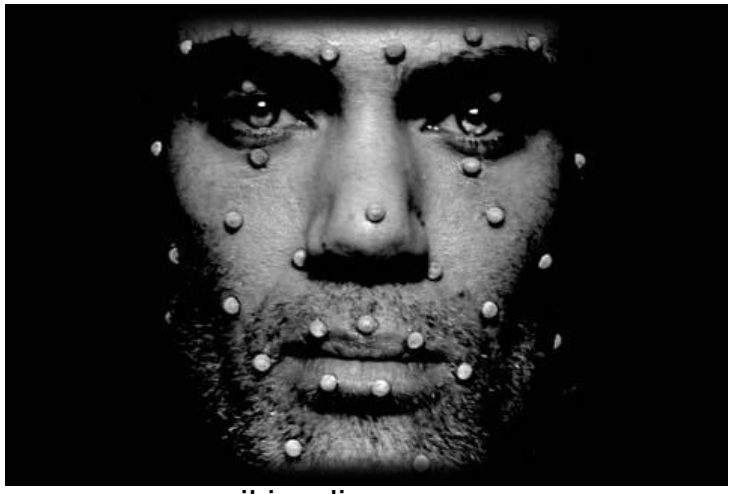
Special effects: motion capture



www.fxguide.com

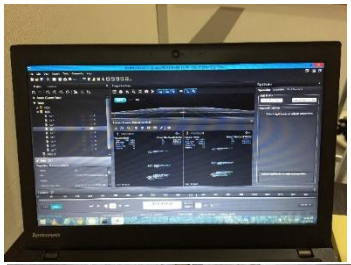


www.davidbordwell.net



en.wikipedia.org

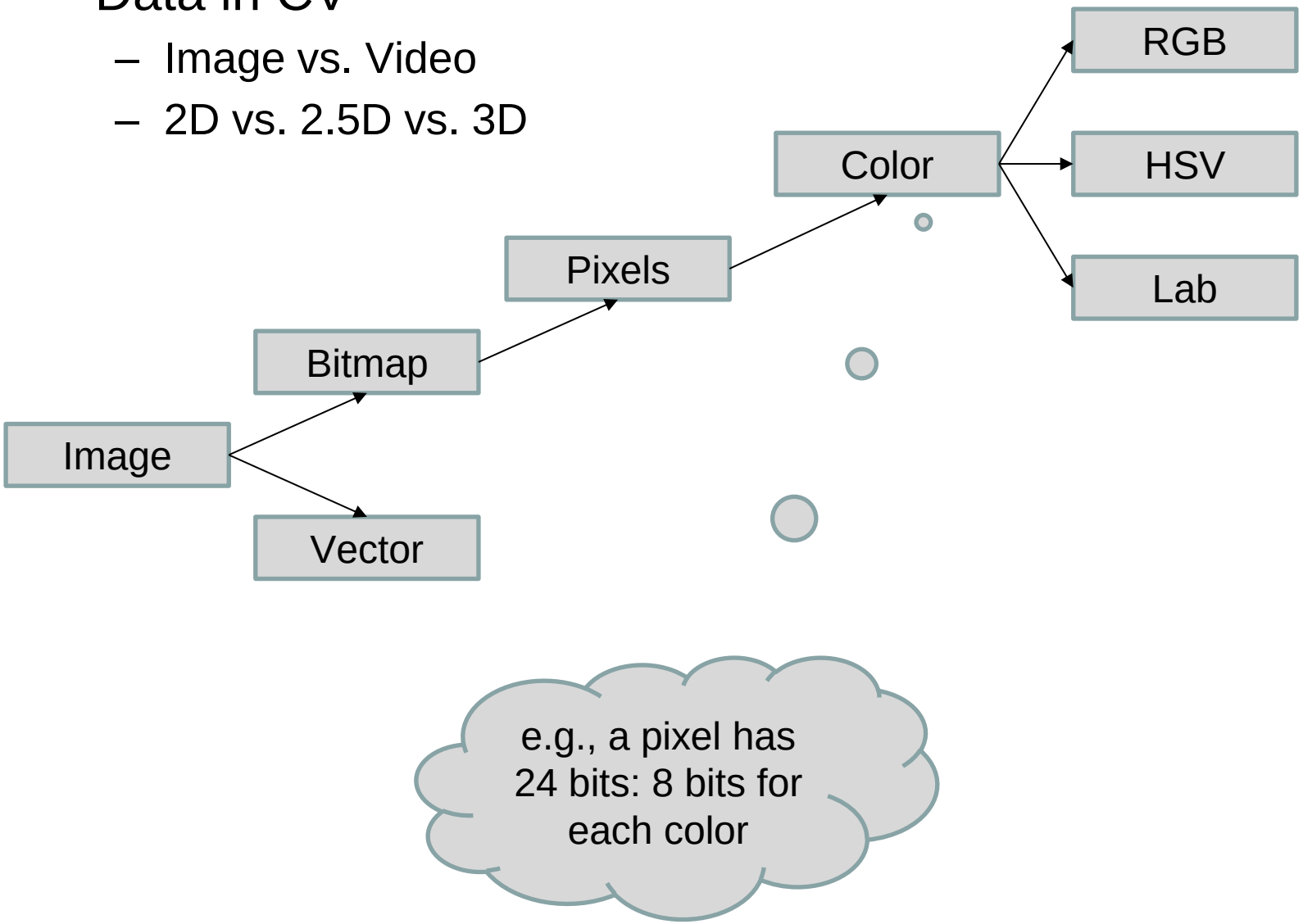






Data

- Data in CV
  - Image vs. Video
  - 2D vs. 2.5D vs. 3D



Type of Images

- Color image
  - e.g. RGB with bit depth = 24
- Grey scale image
  - e.g. bit depth = 8

How many different colors does the 8-bit depth can represent ?



RGB to Grey

Edge detection

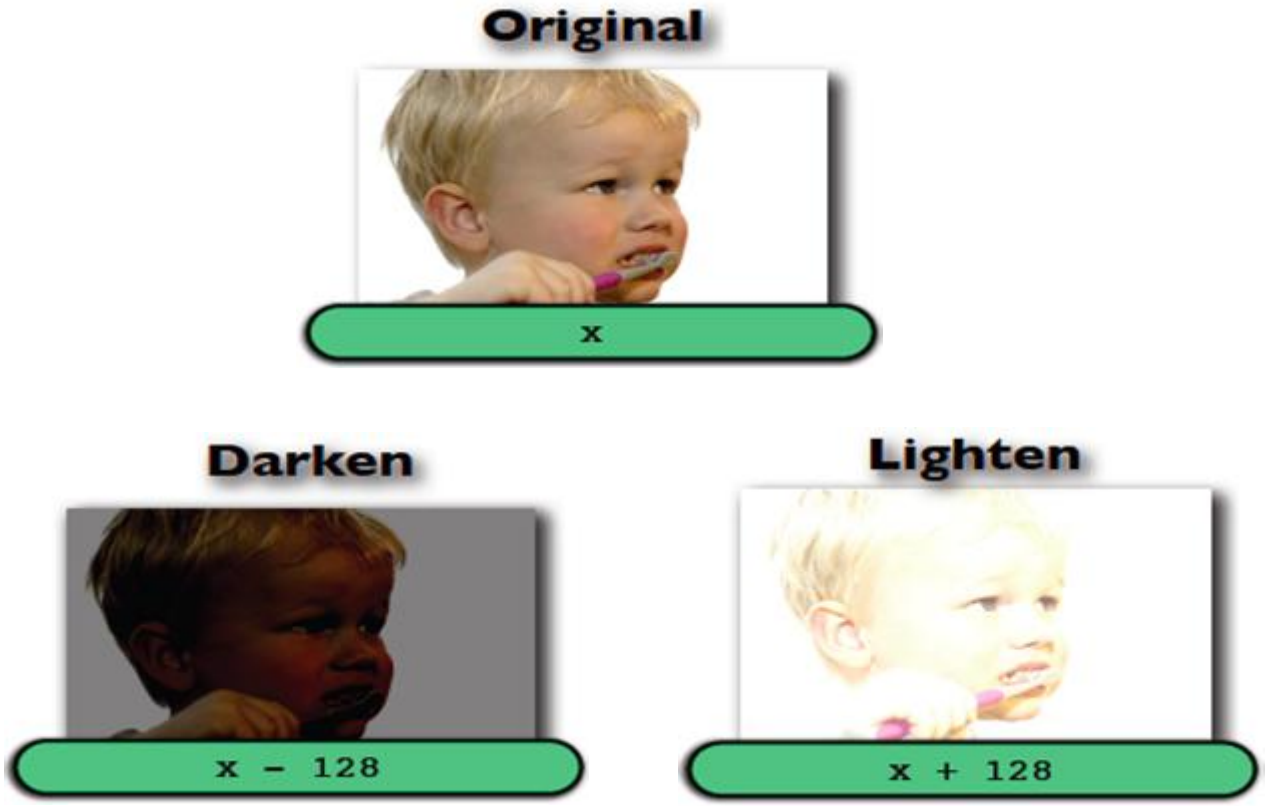
ICDAR2011 database

Processing Operators

- Image processing
  - is the **first stage** in computer vision
  - is the way to process an image into a form suitable for **further analysis**

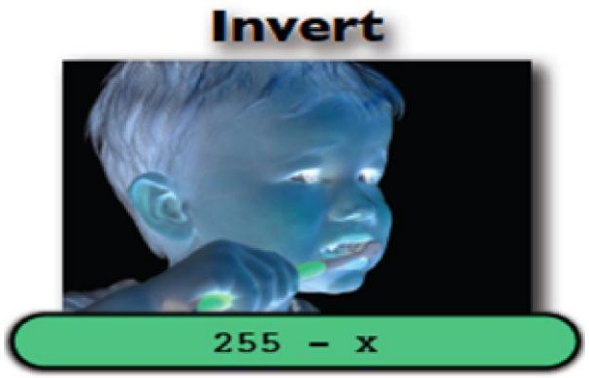
- 2 types
  - Point operators
    - To manipulate each pixel **independently** of its neighbors
  - Neighborhood operators
    - To manipulate each pixel **dependently** of a small number of its neighboring pixels
    - e.g. Linear filtering

Point Operators



Point Operators (cont.)





Point Operators (cont.)



Why can they manage the contrast ?





Point Operators (cont.)

Original



Gamma Correction

Nonlinear
Lower Contrast



Nonlinear
Raise Contrast



Convolutional Operation

An important layer in **CNN**

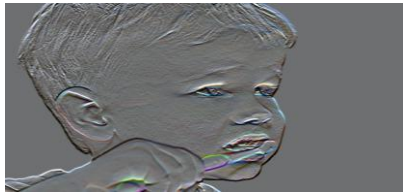




X-Edge Detection



Y-Edge Detection



Edge Detection



Wrapped



Noise



Sharpened Image

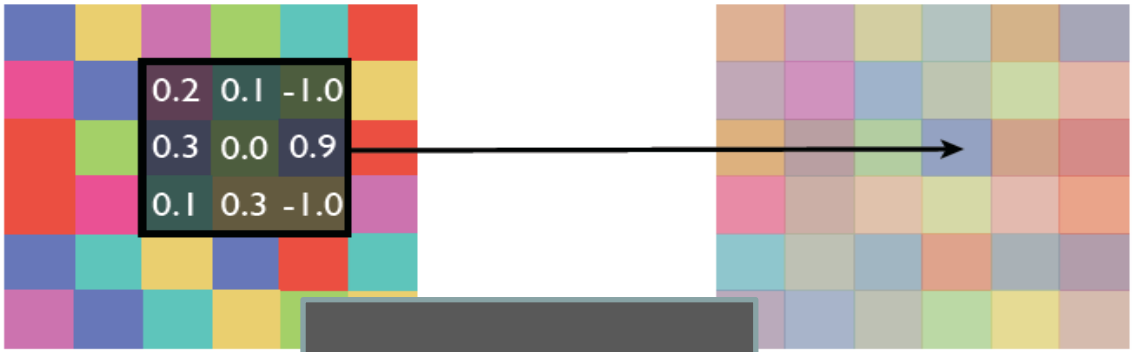


Blurred Image



Nigel Chapman and Jenny Chapman, Digital Multimedia

Convolution



$$F = \begin{bmatrix} 0.2 & 0.1 & -1.0 \\ 0.3 & 0.0 & 0.9 \\ 0.1 & 0.3 & -1.0 \end{bmatrix}$$

$$I' = F * I$$

Edge of the image

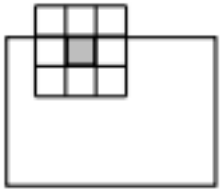


Figure 3.3: A mask at the edge of an image

- At the boundary of an image, the mask won't have enough information to be calculated
- There are different approaches to solve this problem
  - **Ignore the edge**, Masking is applied only an existing pixel which lies fully within the image. Bigger mask would produce a lost of information by this method
  - **Pad** with Zeros, Adding up zero on the missing values around the edge. Sometimes, it introduces artifact around the edges
  - **Pad** with duplication value from the corresponding boundary

Clipping and Normalization

$$y = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \le x \le 255 \\ 255 & \text{if } x > 255 \end{cases}$$

- Averaged result which value is more than 255 or less than 0
  - The application should clip the value into displayable range

$$K = \frac{1}{3 \cdot 3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- Normalized kernel
  - Dividing each element in the kernel by the sum of all elements in the kernel.
    - Sum of all values in mask must be added up to “1”
    - Normalization ensures that the pixel values in output image are of the same relative magnitude as those in the input image

Activity 1.1
Convolution Operation
Padding with duplicates

Sharpen Mask

Size
Values

0	-1	0
-1	5	-1
0	-1	0



Original

Sharpen Filter

Result



7	6	5	5	6	7
6	4	3	3	4	6
5	3	2	2	3	5
5	3	2	2	3	5
6	4	3	3	4	6
7	6	5	5	6	7

A	8	6	**F**	8	9
8	2	1	1	2	**E**
6	1	0	0	1	6
B	1	0	0	1	**D**
8	2	1	1	2	8
9	8	6	**C**	8	9

Have you noticed any pattern or significant changes in the filtered result? If yes, where has it been effected?

Possible result of convolution

- These kind of filtering effect can be:
  - Sharpen
  - Emboss
  - Edge-Detect
  - Smooth
  - Motion-Blur and other effects

To Submit

- Individual work (DON'T FORGET to put your name)
  - Activity 1.1: Paper-based

- Group work (DON'T FORGET to put your members' names)
  - Use your own picture
  - Add 1 more kernel with any other effects and bigger size kernel e.g., 5x5
  - submit .ipynb